

Quantum Coherent Beating in Polar Disubphthalocyanine-Fullerene Cocrystals for Ultrafast Photothermal Conversion

Corresponding Author: Professor Cheng Zhang

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

Chen and co-authors reported a kind of DiSubPc-fullerene cocrystal featuring remarkable photothermal conversion performance, and the underlying mechanism has been thoroughly studied. The new quantum-enhanced strategy unexpectedly and significantly boosts the photothermal conversion and the temperature finally reach as high as 242 degree within such a limited time. In my impression, enhancing the absorbance or the nonradiative decay rate are the two main approaches discussed for improving photothermal conversion. I believe this new strategy with such remarkable enhancement will offer a new sight for the design photothermal conversion materials. Overall, this work is of high quality, and the discussion and viewpoint are solidly supported by experimental and theoretical evidence. In my opinion, this paper is suitable for publication in Nature Photonics after the following aspects further clarified.

1. The evidence of the instability of 4DiSubPc should be provided.
2. In the photothermal conversion experiments, more detailed information should be provided regarding the amount of samples used and the size of the quartz groove, as they are significantly closed to the temperature upon photothermal conversion.
3. The crystals showed excellent photothermal stability during the photothermal conversion experiment. The authors should include the UV-vis-NIR absorption of crystals to discuss the selection of the light source wavelength for photothermal conversion experiment.
4. The PL spectra of 2DiSubPc, 2DiSubPc-C60 and 2DiSubPc-C70 (fig 5a-c) are quite different from each other. While the authors assigned the PL of 2DiSubPc-C60 to the residuals of 2DiSubPc, the PL of 2DiSubPc-C70 was attributed to the emission of C70. This should be further elaborated, together with the PL spectra measured from the solutions (fig 2d,e). I suspect the PL intensities should be very low for 2DiSubPc-C60 and 2DiSubPc-C70. How to exclude the influence of re-absorption in crystal samples? How to distinguish energy transfer from electron transfer in the 2DiSubPc composites?
5. It would be better to measure the PL spectra (maybe also PL lifetimes) from pristine C70 with their own set up, and compare the results with fig 5c.
6. The oscillations in fig 5f are very interesting, which are attributed to the coherent quantum beating of high spin excited states by the authors. Have the authors measured Fs-TAS on the 2DiSubPc and 2DiSubPc-C60 samples? Did they observe similar oscillations? The ultrafast quantum coherent beating is observed at 17.6 GHz. Why is this frequency a bit higher (in my impression) compared with previous reports on exciton quantum beating in organic crystals? In addition, is the transient absorption band in fig 5e associated with TT states?
7. The amplitudes of MPL should be added to the y axis in fig 5g-i, or at least mentioned in the caption/text. Why is the MPL in 2DiSubPc-C70 thermally activated, i.e., it becomes more obvious with the increase of temperature.
8. Is the PL lifetime of 2DiSubPc-C70 also temperature-dependent?
9. Overall, I agree that the observed MPL provides strong evidence for the existence of high spin excited states in 2DiSubPc-C70. However, the explanations, especially for the crucial role of TT states, should be provided in more detail. If possible, the control experiment of MPL from pristine C70 is needed.
10. Error bars should be included for the fittings in fig S11. Can the authors obtain E_a for pristine C70?

Reviewer #2

(Remarks to the Author)

In this manuscript, Chen et al. report the fabrication of donor-acceptor cocrystals by assembling a dumbbell-shaped disubphthalocyanine (2DiSubP) with C60 and C70, respectively, their photothermal property, and investigation on the photothermal mechanism. Their host-guest interactions, crystal structures, photothermal conversion, and photothermal mechanism have been investigated using a variety of characterization techniques, which provide abundant experimental results. The different host-guest interactions between 2DiSubP and C60 or C70 led to a dramatic difference in molecular packing in the donor-acceptor cocrystals, with one being centrosymmetric but the other noncentrosymmetric, which is quite interesting and a rare case. The photothermal performance is impressive. A key contribution of this study is the elucidation of the unique photothermal mechanism of the cocrystals of 2DiSubPc-C70, which can provide useful guidance for the design of high performance photothermal materials. I recommend acceptance for publication in Nature Photonics after some issues are addressed. Specific comments:

1. The photothermal property observed for 2DiSubPc-C70 was attributed to quantum coherent beating, which is the key point of this study. Therefore, it must be confirmed that it is the mechanism that dictates the different photothermal processes of 2DiSubPc-C60 and 2DiSubPc-C70. It might also be explained by other reasons, for example, due to the different host-guest interaction behavior between C60 and C70 arising from their different shape and structure. A deeper investigation to confirm the proposed mechanism, maybe with more control experiences and characterization, is suggested.

2. Toluene molecules were found to co-crystallize with 2DiSubPc-C60 but such a phenomenon was not observed for the cocrystals of 2DiSubPc-C70. The difference might also result in their different photothermal performance. It should be checked.

3. The study on the photothermal conversion should be strengthened.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

Following the authors' careful revision, the comments I raised have been resolved. Based on my assessment, the manuscript could be recommended for acceptance.

Reviewer #2

(Remarks to the Author)

I have checked the revision and found my comments have been properly addressed. I recommend publication of the revised manuscript.

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Point-to-Point **Answers** to Reviewer 1

Overall Comments. Chen and co-authors reported a kind of DiSubPc-fullerene cocrystal featuring remarkable photothermal conversion performance, and the underlying mechanism has been thoroughly studied. The new quantum-enhanced strategy unexpectedly and significantly boosts the photothermal conversion and the temperature finally reach as high as 242 degree within such a limited time. In my impression, enhancing the absorbance or the nonradiative decay rate are the two main approaches discussed for improving photothermal conversion. I believe this new strategy with such remarkable enhancement will offer a new sight for the design photothermal conversion materials. Overall, this work is of high quality, and the discussion and viewpoint are solidly supported by experimental and theoretical evidence. In my opinion, this paper is suitable for publication in Nature Photonics after the following aspects further clarified.

=> We sincerely appreciate the reviewer's insightful comments and valuable advice. The revision has significantly improved the quality of our manuscript.

1. The evidence of the instability of 4DiSubPc should be provided.

=> The half-life ($T_{1/2}$) of **4DiSubPc** was evaluated in a 10^{-5} M toluene solution under ambient conditions to simulate the extended crystal growth process. Based on UV-vis absorption spectral analysis (Figure R1), the $T_{1/2}$ was determined to be 76 hours. These modifications were incorporated into the "Crystal Packings" section and highlighted in yellow. Figure R1 was also provided in the Supporting Information.

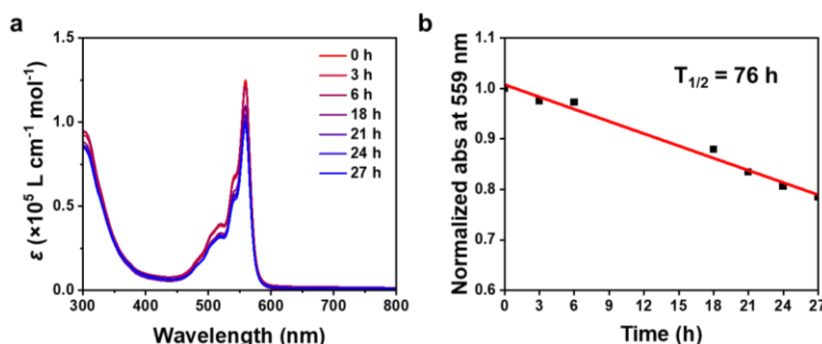


Figure R1. Solution stability evaluation of **4DiSubPc**. (a) time-dependent UV-vis absorption spectra of **4DiSubPc** in a 10^{-5} M toluene solution under ambient conditions. (b) Temporal change in absorbance at 559 nm, with an estimated half-life of 76 hours.

2. In the photothermal conversion experiments, more detailed information should be provided regarding the amount of samples used and the size of the quartz groove, as they are significantly closed to the temperature upon photothermal conversion.

=> The photothermal conversion performance were evaluated under 550 nm laser irradiation (0.5 W cm $^{-2}$) using 9 mg of crystal samples placed in a groove (5 mm $\times$ 5 mm $\times$ 1 mm) at the center of a 7 g quartz plate (50 mm $\times$ 35 mm $\times$ 2 mm). These experimental details were added to the "Methods" section and highlighted in yellow.

3. The crystals showed excellent photothermal stability during the photothermal conversion experiment. The authors should include the UV-vis-NIR absorption of

crystals to discuss the selection of the light source wavelength for photothermal conversion experiment.

=> The UV-vis-NIR absorption spectra of single crystals of C₇₀, C₆₀, and ²DiSubPc, as well as the ²DiSubPc-C₇₀ and ²DiSubPc-C₆₀ cocrystals, were collected under an optical microscope using a white-light source and the reflectance signal from a blank region was used as the reference, which is called micro-area reflectance spectra (*Sci. Adv.* **11**, eadu0298 (2025). *Nat. Commun.* **16**, 9158 (2025)). The results are shown in Figure R2 and also provided in the Supporting Information. Although the cocrystals showed broad absorption bands extending up to 700 nm, a 550 nm light source was selected for photothermal conversion measurements based on two considerations: first, all samples show strong absorption at this wavelength; second, efficient quantum coherent beating between the singlet and (TT)¹ states in ²DiSubPc-C₇₀ has been observed under 530 nm excitation, suggesting potential relevance. Related discussions were included in the “Photothermal conversion” section and highlighted in yellow.

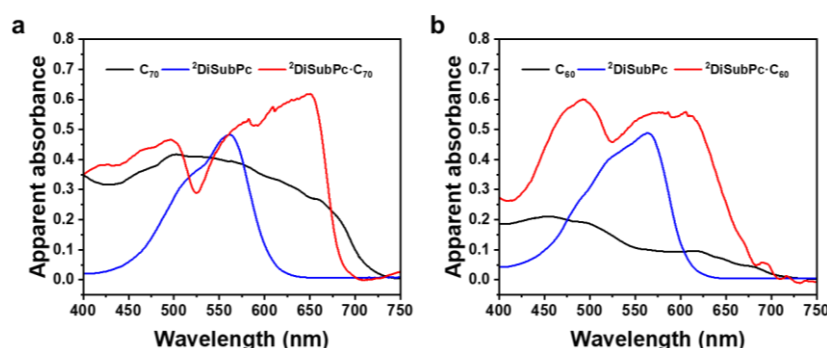


Figure R2. Apparent absorption spectra. (a) Comparison of UV-vis-NIR absorption of C₇₀ and ²DiSubPc single crystals with ²DiSubPc-C₇₀ cocrystals. (b) Comparison of UV-vis-NIR absorption of C₆₀, and ²DiSubPc single crystals with ²DiSubPc-C₆₀ cocrystals.

4. The PL spectra of 2DiSubPc, 2DiSubPc-C60 and 2DiSubPc-C70 (fig 5a-c) are quite different from each other. While the authors assigned the PL of 2DiSubPc-C60 to the residuals of 2DiSubPc, the PL of 2DiSubPc-C70 was attributed to the emission of C70. This should be further elaborated, together with the PL spectra measured from the solutions (fig 2d,e). I suspect the PL intensities should be very low for 2DiSubPc-C60 and 2DiSubPc-C70. How to exclude the influence of re-absorption in crystal samples? How to distinguish energy transfer from electron transfer in the 2DiSubPc composites?

=> Indeed, the PL intensities of the cocrystals are significantly low, which can be attributed to the strong host-guest interactions as evidenced by the titration experiments in Figures 2d-e of the manuscript. To mitigate the effects of re-absorption, we therefore employed a micro-area spectroscopy system for PL measurements, with both excitation and collection performed from the same side of the crystals.

=> The PL peaks of ²DiSubPc-C₆₀ are clearly derived from ²DiSubPc, as their positions remain almost unchanged compared to those of pristine ²DiSubPc. To determine the origin of the PL signal in ²DiSubPc-C₇₀, we measured the PL spectrum of C₇₀, as shown in Figure R3. The PL peak of ²DiSubPc-C₇₀ (731 nm) aligns closely

with that of pristine C₇₀ (760 nm) but exhibits a slight blue shift. This shift can be ascribed to the weakened self-aggregation of C₇₀ and reduced π -conjugation resulting from asymmetric π - π interactions with ²DiSubPc, which also accounts for the polarization observed in the ²DiSubPc-C₇₀ cocrystal. If a new charge-transfer (CT) state formed, the PL peak would have displayed a red shift relative to both ²DiSubPc and C₇₀. Therefore, the PL emission of ²DiSubPc-C₇₀ originates from C₇₀. Figure R3 was also added in the Supporting Information, cited in the manuscript and highlighted in yellow.

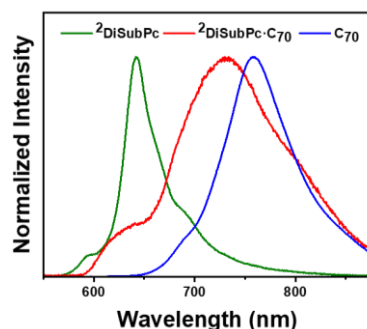


Figure R3. PL Comparison of ²DiSubPc, C₇₀, and ²DiSubPc-C₇₀ crystals.

5. It would be better to measure the PL spectra (maybe also PL lifetimes) from pristine C₇₀ with their own set up, and compare the results with fig 5c.

=> The PL spectra of C₇₀ were measured and are shown in Figure R3, which is also included in the Supporting Information. This comment is closely related to the previous one. Besides, as shown in Figure R4, the PL lifetime of C₇₀ at room temperature was determined to be 0.86 ns, significantly shorter than the 4.96 ns lifetime observed in ²DiSubPc-C₇₀ cocrystals, further reaffirming the appearance of alternative non-radiative processes in the cocrystal finally resulting in the photothermal conversion occurring in ²DiSubPc-C₇₀ cocrystals. Figure R4 was also added in the Supporting Information, cited in the manuscript and highlighted in yellow.

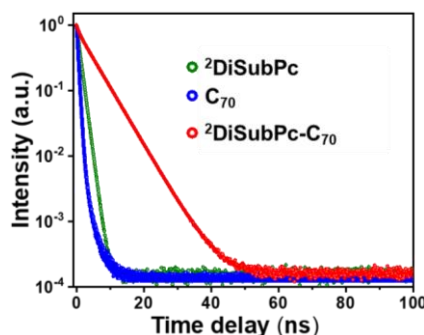


Figure R4. PL lifetime Comparison of ²DiSubPc, C₇₀, and ²DiSubPc-C₇₀ crystals at 300 K.

6. The oscillations in fig 5f are very interesting, which are attributed to the coherent quantum beating of high spin excited states by the authors. Have the authors measured Fs-TAS on the 2DiSubPc and 2DiSubPc-C₆₀ samples? Did they observe similar oscillations? The ultrafast quantum coherent beating is observed at 17.6 GHz. Why is

this frequency a bit higher (in my impression) compared with previous reports on exciton quantum beating in organic crystals? In addition, is the transient absorption band in fig 5e associated with TT states?

=> We performed fs-TA measurements on **²DiSubPc** and **²DiSubPc-C₆₀**, as presented in Figure R5. As shown in Figures R5a and R5c, oscillatory behavior is also observed in the **²DiSubPc-C₆₀** sample (6.0 GHz), though at a frequency three times lower than that of **²DiSubPc-C₇₀** (17.6 GHz). In contrast, no oscillatory signal is detected in **²DiSubPc** alone; instead, only singlet fission from the singlet excited state to the triplet-pair state (¹TT) is observed, as illustrated in R5b and R5d. The singlet fission process ($S_1 \rightarrow ^1TT$) was fitted with a time constant of 2.5 ps. This singlet fission step is followed by relaxation of the triplet pair with a time constant of 300 ps producing the conversion likely through a conical intersection of the excitation into heat (i.e., also producing a residual emission from C₇₀). These results strongly indicate that the oscillation occurs exclusively in the presence of C₆₀ or C₇₀. The rapid energy exchange between **²DiSubPc** and C₆₀/C₇₀ gives rise to the quantum beating signal observed in the kinetics. The beating frequency is directly correlated with the energy transfer rate between the two moieties, which in turn reflects their binding affinity. The fs-TA data further reveal a stronger oscillation signal in the presence of C₇₀, indicating superior binding interaction, consistent with its enhanced photothermal performance. The transient absorption band corresponds to the excited-state absorption of the ¹TT state. Between 0 and 20 ps in Figure R1b, the TA profile closely resembles that of the **²DiSubPc-C₇₀** sample, confirming the assignment of the TA signal.

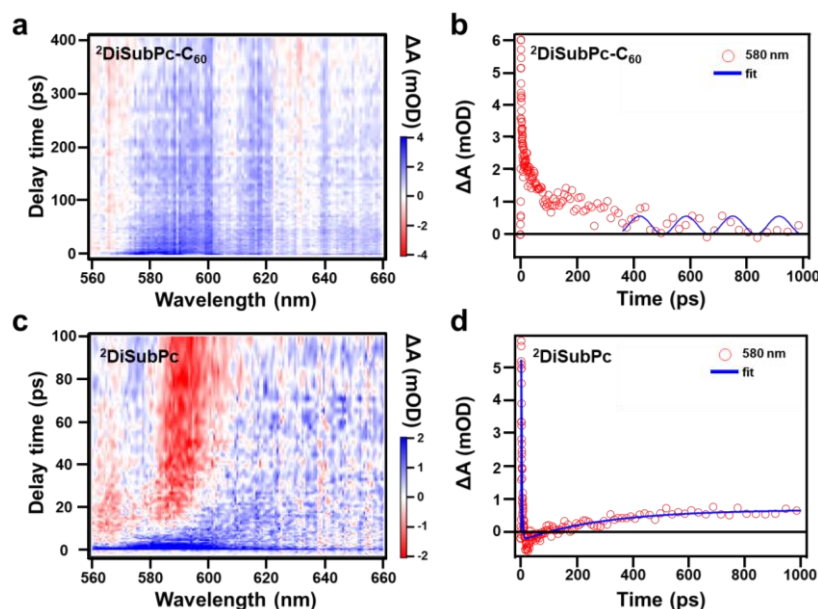


Figure R5. Pseudo color plot of the transient absorption spectra of (a) **²DiSubPc-C₆₀** and (c) **²DiSubPc**. Kinetics at 580 nm of (b) **²DiSubPc-C₆₀** and (d) **²DiSubPc**.

=> Due to this fs-TA results are super important to elucidate our proposed photothermal conversion mechanism, we incorporated the corresponding data in Figure 5. Relative discussions were also added in the “Photothermal Mechanisms” section and highlighted in yellow.

7. The amplitudes of MPL should be added to the y axis in fig 5g-i, or at least mentioned in the caption/text. Why is the MPL in 2DiSubPc-C70 thermally activated, i.e., it becomes more obvious with the increase of temperature.

=> The amplitudes of MPL spectra were clearly indicated in the figure caption.

=> As shown in Figure R6 and Table R1, temperature-dependent PL lifetime measurements were conducted. With decreasing temperature, molecular thermal vibrations are suppressed, leading to an increased contribution of C₇₀ emission and a reduction in quantum beating energy transfer, which ultimately results in the disappearance of the MPL signal.

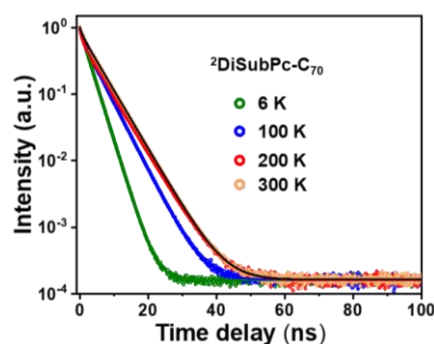


Figure R6. Temperature-dependent PL lifetime spectra of ²DiSubPc-C₇₀ cocrystals.

Table R1. Temperature-dependent PL lifetime of ²DiSubPc-C₇₀ cocrystals.

² DiSubPc-C ₇₀	6 K	100 K	200 K	300 K
A ₁	54.8%	39.9%	19.3%	15.5%
τ ₁ (ns)	0.73	0.71	0.71	0.73
A ₂	45.2%	60.1%	80.7 %	84.5%
τ ₂ (ns)	3.72	4.20	4.62	4.96

8. Is the PL lifetime of 2DiSubPc-C70 also temperature-dependent?

=> Yes, the PL lifetime of ²DiSubPc-C₇₀ cocrystals is temperature-dependent and the results are shown in Figure R6 and Table S3. Both figure and table were also added in the Supporting Information.

9. Overall, I agree that the observed MPL provides strong evidence for the existence of high spin excited states in in 2DiSubPc-C70. However, the explanations, especially for the crucial role of TT states, should be provided in more detail. If possible, the control experiment of MPL from pristine C70 is needed.

=> First, to elucidate the formation of the TT state, we conducted fs-TA measurements; details are provided in Comments 1–6. Second, the MPL spectra of pristine C₇₀ were acquired and are displayed in Figure R7 (included in the Supporting Information). The absence of MPL signals from pristine C₇₀ indicates that the MPL signals observed in the ²DiSubPc-C₇₀ cocrystals originate not from the individual components but from their interactions in the excited state. The rapid energy exchange and the prolonged

excited-state lifetime allow for the spin-mixing across sublevels of $^1(\text{TT})$ state, which is sensitive to external magnetic field. Corresponding discussions were added to the manuscript and highlighted in yellow.

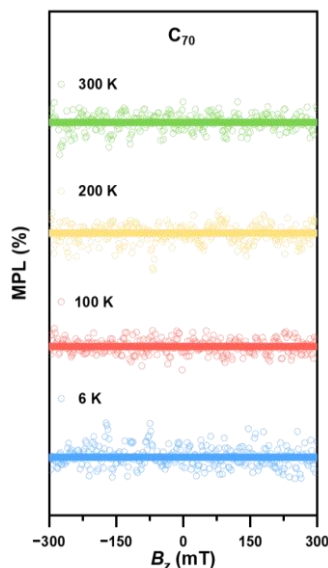


Figure R7. The magneto-photoluminescence spectra at different temperatures of C_{70} .

10. Error bars should be included for the fittings in fig S11. Can the authors obtain E_a for pristine C_{70} ?

=> The error bars were added in Figure S17.

=> As shown in Figure R8, We checked the temperature-dependent PL spectra of C_{70} and the E_a for the nonradiative pathways of excited states was evaluated to be 35 meV. Figure R8 was also added in the Supporting Information, cited in the manuscript and highlighted in yellow.

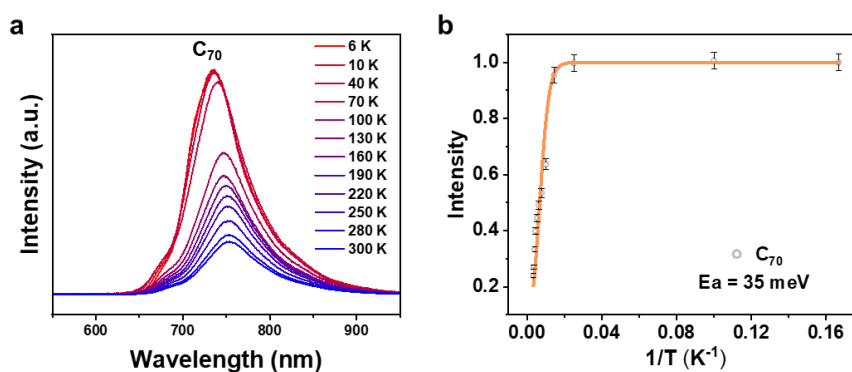


Figure R8. The temperature-dependent PL spectra from 6 K to 300 K of C_{70} and the corresponding integral of fluorescence intensity (black dots) employed for linear fitting, with the fitting curve and obtained activation energy displayed.

Point-to-Point **Answers** to Reviewer 2

Overall Comments. In this manuscript, Chen et al. report the fabrication of donor-acceptor cocrystals by assembling a dumbbell-shaped disubphthalocyanine (2DiSubP) with C60 and C70, respectively, their photothermal property, and investigation on the photothermal mechanism. Their host-guest interactions, crystal structures, photothermal conversion, and photothermal mechanism have been investigated using a variety of characterization techniques, which provide abundant experimental results. The different host-guest interactions between 2DiSubP and C60 or C70 led to a dramatic difference in molecular packing in the donor-acceptor cocrystals, with one being centrosymmetric but the other noncentrosymmetric, which is quite interesting and a rare case. The photothermal performance is impressive. A key contribution of this study is the elucidation of the unique photothermal mechanism of the cocrystals of 2DiSubPc-C70, which can provide useful guidance for the design of high performance photothermal materials. I recommend acceptance for publication in Nature Photonics after some issues are addressed.

=> We sincerely thank the reviewer for the positive feedback and valuable suggestions. The reviewer's observations align closely with the previous assessment. We believe that the revisions have substantially enhanced the quality of the manuscript.

1. The photothermal property observed for 2DiSubPc-C70 was attributed to quantum coherent beating, which is the key point of this study. Therefore, it must be confirmed that it is the mechanism that dictates the different photothermal processes of 2DiSubPc-C60 and 2DiSubPc-C70. It might also be explained by other reasons, for example, due to the different host-guest interaction behavior between C60 and C70 arising from their different shape and structure. A deeper investigation to confirm the proposed mechanism, maybe with more control experiences and characterization, is suggested.

=> **²DiSubPc** exhibit strong host-guest interactions with both C₆₀ and C₇₀, which we believe underlies the quantum coherent beating observed in **²DiSubPc-C₇₀** cocrystals. However, we should figure out the reason why the photothermal conversion performance of **²DiSubPc-C₆₀** is inferior to **²DiSubPc-C₇₀**. To address this discrepancy, we carried out extensive experiments to understand the mechanisms. In the revised manuscript and Supporting Information, we now present fs-TA spectra of **²DiSubPc-C₆₀** and **²DiSubPc**, PL spectra of C₇₀, temperature-dependent PL lifetime of **²DiSubPc-C₇₀**, as well as temperature-dependent MPL data of C₇₀ (Figure R9). These results offer clear evidence for the formation of the ¹(TT) state and quantum beating in both cocrystal systems. Notably, the beating frequency in **²DiSubPc-C₆₀** is three times lower than that in **²DiSubPc-C₇₀**, implying weaker binding interactions, which is consistent with its lower photothermal performance.

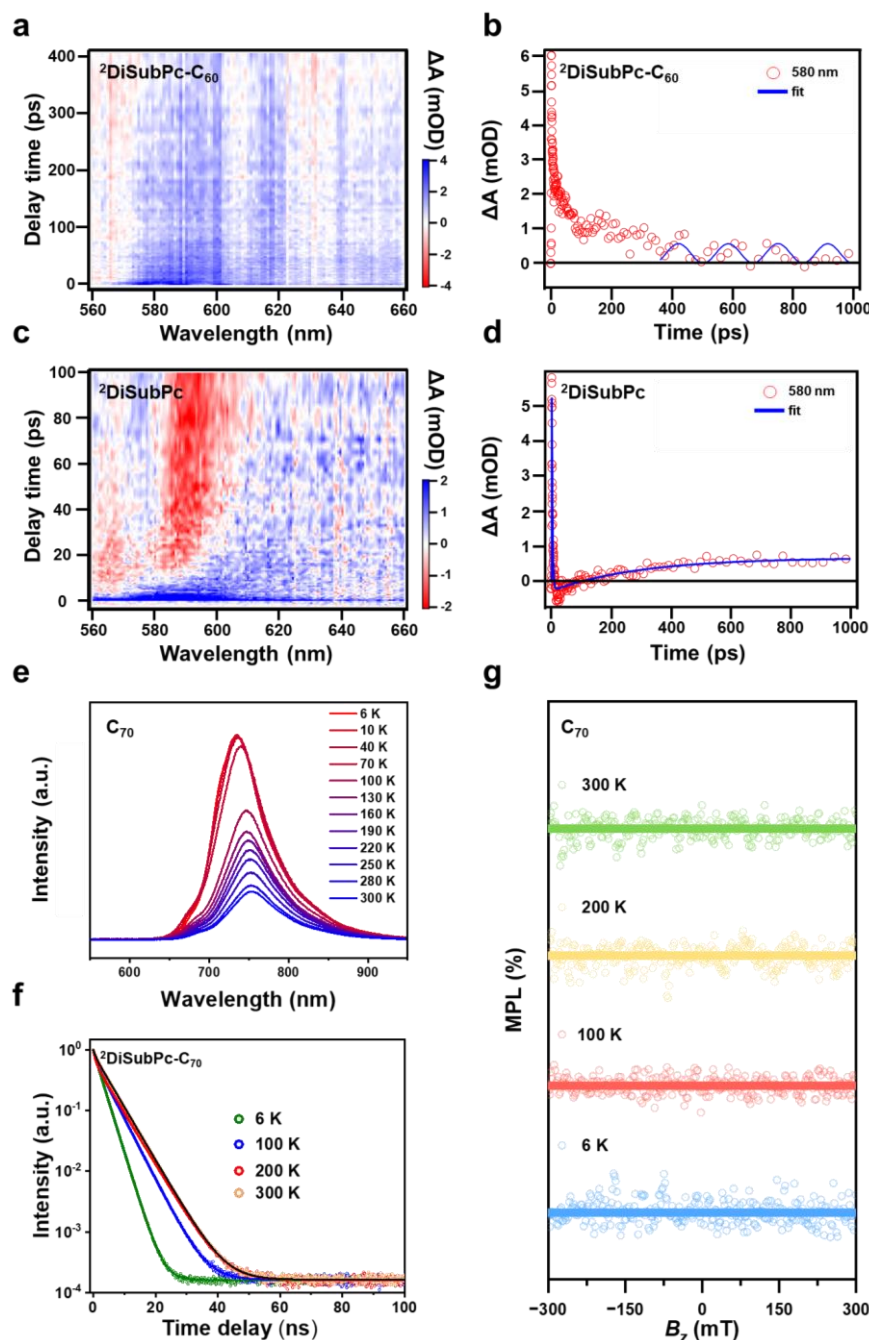


Figure R9. Pseudo color plot of the transient absorption spectra of (a) ²DiSubPc-C₆₀ and (c) ²DiSubPc. Kinetics at 580 nm of (b) ²DiSubPc-C₆₀ and (d) ²DiSubPc. (e) The temperature-dependent PL spectra from 6 K to 300 K of C₇₀. (f) Temperature-dependent PL lifetime spectra of ²DiSubPc-C₇₀ cocrystals. (g) The magneto-photoluminescence spectra at different temperatures of C₇₀.

2. Toluene molecules were found to co-crystallize with 2DiSubPc-C₆₀ but such a phenomenon was not observed for the cocrystals of 2DiSubPc-C₇₀. The difference might also result in their different photothermal performance. It should be checked.

=> We tried tremendous approaches to obtain solvent-free cocrystals of ²DiSubPc-C₆₀ and ²DiSubPc-C₇₀. However, ²DiSubPc-C₆₀ cocrystals consistently incorporated toluene. To clarify the role of toluene in the cocrystal structure, we performed void

analysis using Mercury software. The results indicated that the void volumes in $^2\text{DiSubPc-C}_{60}$ and $^2\text{DiSubPc-C}_{70}$ cocrystals accounted for 3.5% and 12% of the unit cell volume, respectively (Figure R10). However, upon removal of toluene from the $^2\text{DiSubPc-C}_{60}$ structure, the voids increased significantly from 3.5% to 25.6%, suggesting that the resulting voids are too large to sustain the original packing arrangement, likely leading to structural collapse. At the same time, the voids in $^2\text{DiSubPc-C}_{70}$ cocrystals is insufficient to accommodate toluene molecules.

=> Besides, as shown in Figure R11, the physical blend of $^2\text{DiSubPc}$ and C_{60} exhibited a decomposition temperature of 436 °C, consistent with that of pristine $^2\text{DiSubPc}$ (Figure S1), confirming the good thermal stability of both components. However, the mass fraction of toluene eliminated from $^2\text{DiSubPc-C}_{60}$ cocrystals was calculated as 12.9%, occurring at 450 °C, indicating that $^2\text{DiSubPc}$ decompose upon heating during toluene removal. This observation is consistent with the fact that the $^2\text{DiSubPc-C}_{60}$ cocrystals remained stable at the high equilibrium temperature of 211 °C, which far more exceeds the boiling temperature of toluene. This also serves as evidence of strong intermolecular interactions involving toluene molecules. Therefore, toluene is essential for stabilizing the $^2\text{DiSubPc-C}_{60}$ packings but cannot be incorporated into the $^2\text{DiSubPc-C}_{70}$ structures.

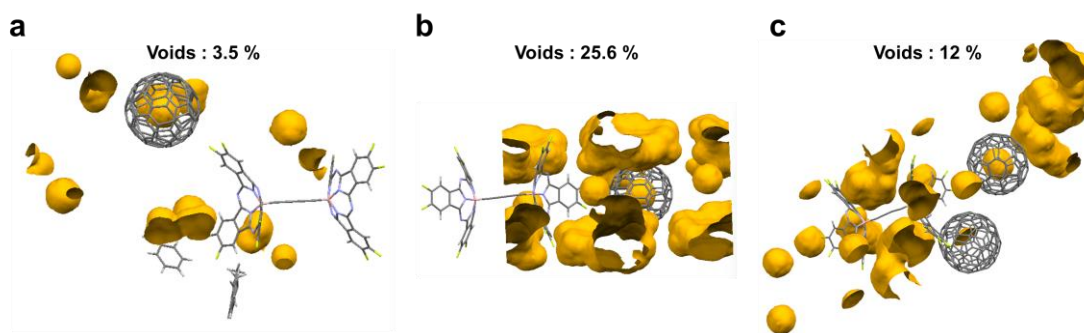


Figure R10. Crystal voids of a) $^2\text{DiSubPc-C}_{60}$, b) $^2\text{DiSubPc-C}_{60}$ without toluenes, and c) $^2\text{DiSubPc-C}_{70}$.

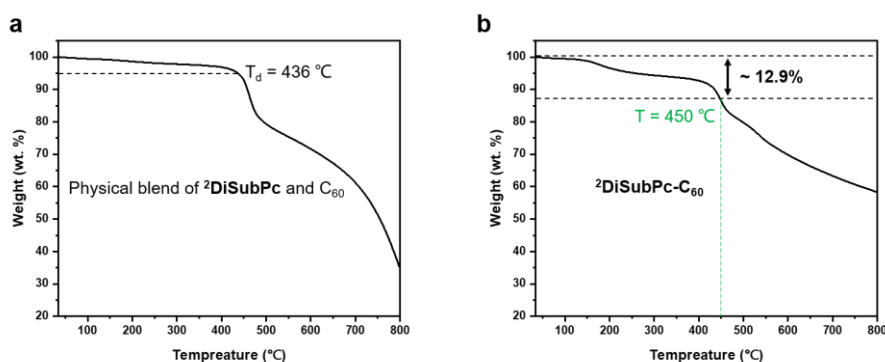


Figure R11. The TGA curves of (a) physical blend of $^2\text{DiSubPc}$ and C_{60} and (b) $^2\text{DiSubPc-C}_{60}$ cocrystals involving toluenes.

=> Figure R10-11 were also included in the Supporting Information. Corresponding discussions were also incorporated into the manuscript and highlighted in yellow.

3. The study on the photothermal conversion should be strengthened.

=> Our investigation on the photothermal conversion is enhanced by focusing on the following three aspects.

=> First, we performed several control experiments to optimize the best irradiation conditions, including wavelength-dependent and intensity-dependent photothermal conversion experiments. As shown in Figure R12a, 550 nm was identified as the most suitable wavelength for our system, achieving the highest temperature, which is consistent with the excitation wavelength to generate the quantum beating. Meanwhile, temperature increase exhibited a linear relationship with light intensity (Figure R12b-c), allowing for fair comparison with other state-of-the-art organic materials after normalization by optical power. Under the optimized conditions, we included pristine C₆₀ and C₇₀ as reference materials, which reached temperatures of 120 and 119 °C within 5 s and equilibrated at 130 and 128 °C with good stability, respectively, evidencing this by infrared thermal images (Figure R12d-f). Figure R12 was also included as Figure 4 in the main text and Figure S19-20 in the Supporting Information. Corresponding discussions were also incorporated into the manuscript and highlighted in yellow.

=> Second, the photothermal conversion mechanisms were thoroughly investigated in accordance with the Reviewers' comments. All comments were adequately addressed, and corresponding revisions were made to the manuscript.

=> Third, the experimental details of photothermal conversion experiments were added to the "Methods" section in the main text.

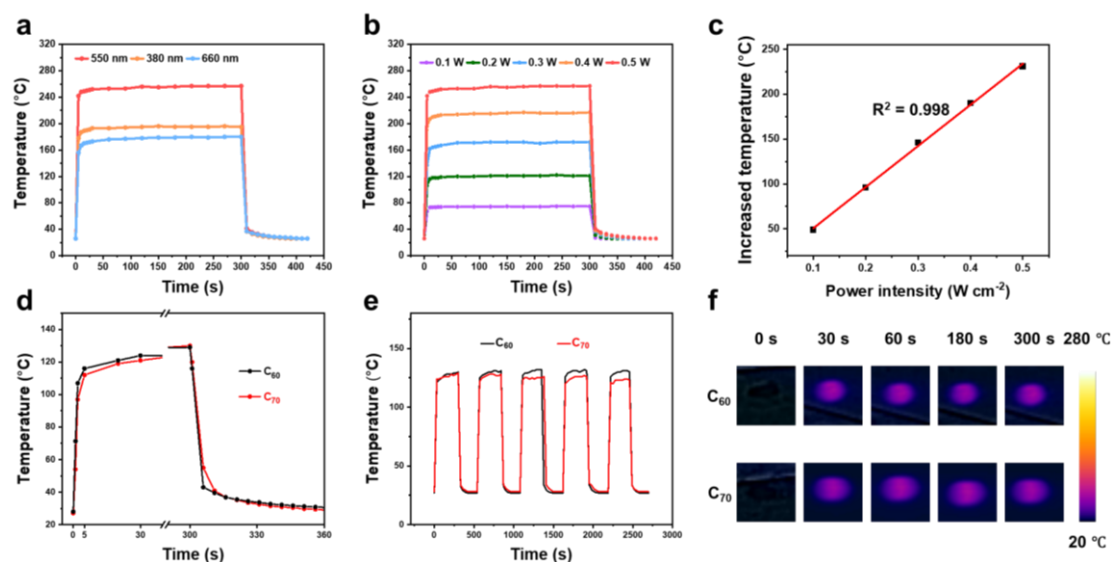


Figure R12. (a) Wavelength-dependent and (b) intensity-dependent photothermal conversion experiments. (c) Linear relationship between light intensity and temperature. (d) Temperature variations, (e) photothermal stability, and (f) infrared thermal images of C₆₀ and C₇₀.

Point-to-Point [Answers](#) to Reviewer 1

Overall Comments. Following the authors' careful revision, the comments I raised have been resolved. Based on my assessment, the manuscript could be recommended for acceptance.

=> We sincerely appreciate the reviewer's recommendation for acceptance. Following the thoughtful suggestions and advices, the quality of our manuscript is largely improved.

Point-to-Point [Answers](#) to Reviewer 2

Overall Comments. I have checked the revision and found my comments have been properly addressed. I recommend publication of the revised manuscript.

=> We sincerely thank the reviewer for the positive feedback. We believe the revised manuscript have meaningfully enhanced its clarity, rigor, and overall impact.